Time Series Analysis

Chapter 1 — Basic statistical concepts

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Chapter 1

Basic statistical concepts

Introduction

Before studying time series proper, this chapter reviews the statistical concepts on which the rest of the course relies: *moments* of a distribution, classical *point estimators*, *Method of Moments* and *Maximum Likelihood Estimation*, sampling distributions, and basic *inference* (confidence intervals and hypothesis tests).

The convention throughout the chapter is $X_1, \ldots, X_n$ are *independent and identically distributed* (i.i.d.) draws from a distribution with density $f(x; \theta)$ parameterised by some unknown θ. We observe a sample and want to learn θ. Time series will later violate the i.i.d. assumption, but the basic estimators introduced here remain the building blocks.

1.1 Moments of a distribution

Definition 1 – Raw and central moments

Let X be a real-valued random variable with density f and mean $\mu = \mathbb{E}[X]$.

- The *raw ℓ-th moment* is $m'_\ell = \mathbb{E}[X^\ell] = \int x^\ell f(x)\, dx$.
- The *central ℓ-th moment* is $m_\ell = \mathbb{E}[(X - \mu)^\ell]$.

Useful quantities.

$$\mu = m'_1, \qquad \sigma^2 = m_2 = \mathrm{Var}[X],$$
$$\text{skewness } S(X) = \frac{m_3}{\sigma^3}, \qquad \text{kurtosis } K(X) = \frac{m_4}{\sigma^4}.$$

Remark – Interpretation

- $S(X)$ measures *asymmetry*. $S = 0$ for any symmetric distribution; $S > 0$ for a right-skewed distribution.
- $K(X)$ measures *tail-heaviness*. The Gaussian has $K = 3$; the *excess kurtosis* $K - 3$ is therefore the standard comparison: $K - 3 > 0$ indicates heavier tails than the Gaussian (leptokurtic).

Example 2 – Moments of the standard normal

For $X \sim \mathcal{N}(\mu, \sigma^2)$ one shows by direct integration or via the moment-generating function:

$$\mathbb{E}[X] = \mu, \quad \mathrm{Var}[X] = \sigma^2, \quad m_3 = 0, \quad m_4 = 3\,\sigma^4.$$

Hence $S(X) = 0$ and $K(X) = 3$.

1.2 Classical estimators and sampling distributions

1.2.1 Sample mean and sample variance

Definition 3 – Empirical estimators

Given i.i.d. data $X_1, \dots, X_n$:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

Proposition 4 – Unbiasedness of $\bar{X}_n$ and S_n^2

For i.i.d. data with finite second moments, $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{E}[S_n^2] = \sigma^2$.

Proof. For the mean: $\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_i \mathbb{E}[X_i] = \mu$ by linearity. For the variance, expand:

$$\sum_i (X_i - \bar{X}_n)^2 = \sum_i X_i^2 - n\,\bar{X}_n^2.$$

Take expectations: $\mathbb{E}[X_i^2] = \mu^2 + \sigma^2$ and $\mathbb{E}[\bar{X}_n^2] = \mu^2 + \sigma^2/n$. Hence $\mathbb{E}[\sum_i (X_i - \bar{X}_n)^2] = n(\mu^2+\sigma^2) - n(\mu^2 + \sigma^2/n) = (n-1)\,\sigma^2$. The $n-1$ denominator in S_n^2 ensures $\mathbb{E}[S_n^2] = \sigma^2$. $\square$

Remark – Why $n-1$?

Replacing the unknown μ by $\bar{X}_n$ in the variance formula "costs" one degree of freedom. Dividing by n would systematically underestimate σ^2.

1.2.2 Sampling distribution under normality

Theorem 5 – Cochran-type results

If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then:

1. $\bar{X}_n \sim \mathcal{N}(\mu,\, \sigma^2/n)$;
2. $\dfrac{(n-1)\,S_n^2}{\sigma^2} \sim \chi^2_{n-1}$;
3. $\bar{X}_n$ and S_n^2 are independent;
4. $\dfrac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t_{n-1}$.

Remark

Result (4) follows from (1)–(3): the numerator is standard normal, the denominator is the square root of an independent χ^2_{n-1} divided by its degrees of freedom — the very definition of Student's t.

1.2.3 Asymptotic results

Theorem 6 – Law of Large Numbers

For i.i.d. data with $\mathbb{E}[X_i] = \mu$:

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \to \infty.$$

Theorem 7 – Central Limit Theorem

For i.i.d. data with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } n \to \infty.$$

Remark – Practical consequence

Even when the X_i are not Gaussian, the sample mean behaves approximately as a normal random variable for large n. The rule of thumb $n \geq 30$ is conservative; in practice the convergence is faster for symmetric, light-tailed distributions and slower for skewed or heavy-tailed ones.

1.3 Estimation methods

1.3.1 Method of Moments (MoM)

Definition 8 – Method of Moments

Let $\theta \in \mathbb{R}^k$. The MoM estimator $\widehat{\theta}_n^{\mathrm{MoM}}$ solves the system of equations

$$\mathbb{E}_\theta[X^p] \;=\; \frac{1}{n}\sum_{i=1}^{n} X_i^p, \qquad p = 1, 2, \ldots, k.$$

That is, theoretical moments are equated to sample moments.

Example 9 – $\mathcal{N}(\mu, \sigma^2)$ via MoM

With $\theta = (\mu, \sigma^2)$ and $k = 2$:

$$\mu = \bar{X}_n,$$
$$\mu^2 + \sigma^2 = \frac{1}{n}\sum_i X_i^2.$$

Subtracting yields $\widehat{\sigma}^2_{\mathrm{MoM}} = \frac{1}{n}\sum(X_i - \bar{X}_n)^2$, which differs from S_n^2 by a factor $n/(n-1)$.

1.3.2 Maximum Likelihood Estimation (MLE)

Definition 10 – Likelihood and MLE

For i.i.d. data with density $f(x;\theta)$:

$$L(\theta) = \prod_{i=1}^{n} f(X_i;\theta), \qquad \ell(\theta) = \log L(\theta).$$

The MLE is $\widehat{\theta}_n^{\mathrm{MLE}} = \arg\max_\theta \ell(\theta)$.

Example 11 – $\mathcal{N}(\mu, \sigma^2)$ via MLE

The log-likelihood is

$$\ell(\mu,\sigma^2) = -\tfrac{n}{2}\log(2\pi\sigma^2) - \tfrac{1}{2\sigma^2}\sum_{i=1}^{n}(X_i-\mu)^2.$$

Setting $\partial\ell/\partial\mu = 0$ gives $\widehat{\mu}_{\mathrm{MLE}} = \bar{X}_n$. Then setting $\partial\ell/\partial\sigma^2 = 0$ gives $\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum(X_i - \bar{X}_n)^2$. So MLE coincides with MoM here, and both are biased for σ^2 (their expectation is $\frac{n-1}{n}\sigma^2$).

Theorem 12 – Asymptotic normality of the MLE

Under regularity conditions (identifiability, smoothness, finite Fisher information $I(\theta_0) > 0$):

$$\sqrt{n}\,(\widehat{\theta}_n^{\mathrm{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0,\, I(\theta_0)^{-1}).$$

Remark – MoM vs. MLE

- MoM is computationally simple — just match moments.
- MLE is *asymptotically efficient*: its variance attains the Cramér–Rao lower bound $I(\theta_0)^{-1}/n$.
- Both estimators are consistent (under regularity).
- For "nice" families (Gaussian, exponential, Poisson), MoM and MLE often coincide or are very close.

1.4 Multivariate data

1.4.1 Joint, marginal, and conditional distributions

Definition 13 – Joint and marginal density

For two random variables X, Y with joint density $f(x, y)$:

$$f_X(x) = \int f(x,y)\,dy, \qquad f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}.$$

The chain rule reads $f(x, y) = f_{Y|X}(y \mid x)\, f_X(x)$.

Definition 14 – Independence

X and Y are *independent* iff $f(x,y) = f_X(x)\, f_Y(y)$ for all x, y.

1.4.2 Covariance and correlation

Definition 15 – Covariance and correlation

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)], \qquad \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}X\,\mathrm{Var}Y}}.$$

Proposition 16 – Sample versions

$$\widehat{\mathrm{Cov}}(X,Y) = \tfrac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})(Y_i-\bar{Y}), \qquad \widehat{\mathrm{Corr}}(X,Y) = \frac{\widehat{\mathrm{Cov}}(X,Y)}{\widehat{\sigma}_X\,\widehat{\sigma}_Y}.$$

The sample correlation lies in $[-1,1]$.

Warning – Correlation captures only linear dependence

$\mathrm{Corr}(X,Y) = 0$ does not imply independence. Counter-example: let X be symmetric around 0 and $Y = X^2$. Then $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$, hence $\mathrm{Cov}(X,Y) = 0$, yet Y is a deterministic function of X.

1.5 Statistical inference

1.5.1 Confidence intervals for the mean

Definition 17 – Confidence interval

A $(1-\alpha)$-confidence interval for μ is a random interval $[\widehat{L}_n, \widehat{U}_n]$ such that $\mathbb{P}(\widehat{L}_n \le \mu \le \widehat{U}_n) = 1-\alpha$ in repeated sampling.

Gaussian sample, known σ (z-interval). Let $X_1,\dots,X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu,\sigma^2)$ with σ known. By Theorem 1.2.2 (1):

$$\mathrm{CI}_{1-\alpha}(\mu) = \Big[\, \bar{X}_n - z_{1-\alpha/2}\,\tfrac{\sigma}{\sqrt{n}},\ \bar{X}_n + z_{1-\alpha/2}\,\tfrac{\sigma}{\sqrt{n}} \,\Big].$$

Gaussian sample, unknown σ (t-interval). Replace σ by S_n and use Student's t (Theorem 1.2.2 (4)):

$$\mathrm{CI}_{1-\alpha}(\mu) = \Big[\, \bar{X}_n - t_{n-1,1-\alpha/2}\,\tfrac{S_n}{\sqrt{n}},\ \bar{X}_n + t_{n-1,1-\alpha/2}\,\tfrac{S_n}{\sqrt{n}} \,\Big].$$

1.5.2 Hypothesis testing

Definition 18 – Test on the mean

Two-sided test: $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. Test statistic

$$T \;=\; \frac{\bar{X}_n - \mu_0}{S_n/\sqrt{n}} \;\sim\; t_{n-1} \text{ under } H_0.$$

Reject H_0 at level α if $|T| > t_{n-1,1-\alpha/2}$, i.e. the p-value $2 \cdot \mathbb{P}(T_{n-1} > |T|) < \alpha$.

Remark – Errors

- Type I error (size of the test): $\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0)$.
- Type II error: $\beta = \mathbb{P}(\text{not reject } H_0 \mid H_1)$.
- *Power* of the test: $1 - \beta$, the probability of correctly rejecting H_0 when H_1 is true.

Example 19 – Numerical illustration

Suppose $n = 50$ Atlanta monthly average temperatures with sample mean $\bar{X}_{50} = 61.2$ and sample standard deviation $S_{50} = 16.8$. Test $H_0 : \mu = 60$ against $H_1 : \mu \neq 60$ at $\alpha = 0.05$.

$$T = \frac{61.2 - 60}{16.8/\sqrt{50}} \approx 0.505, \qquad t_{49,0.975} \approx 2.010.$$

Since $|T| < 2.010$, we do *not* reject H_0; p-value $\approx 2 \cdot \mathbb{P}(t_{49} > 0.505) \approx 0.616$. The corresponding 95% CI is $[61.2 - 4.78,\ 61.2 + 4.78] = [56.4,\ 66.0]$, which contains $\mu_0 = 60$, consistent with the test conclusion.

Warning – Common pitfalls

- "Failing to reject H_0" is *not* "proving H_0".
- A small p-value does not imply a large effect — it only signals statistical significance, not practical importance.
- Multiple testing inflates Type I error: use Bonferroni or FDR corrections when running many tests.

Summary

Quantity	Estimator	Sampling distribution (Gaussian data)
μ	$\bar{X}_n = \frac{1}{n}\sum X_i$	$\mathcal{N}(\mu, \sigma^2/n)$
σ^2	$S_n^2 = \frac{1}{n-1}\sum(X_i - \bar{X}_n)^2$	$(n-1)S_n^2/\sigma^2 \sim \chi^2_{n-1}$
Pivot	$T = (\bar{X}_n - \mu)/(S_n/\sqrt{n})$	t_{n-1}

What to remember.

- Two estimation routes: MoM (match moments) and MLE (maximise likelihood). Both consistent; MLE asymptotically efficient.
- Under normality, the sampling distribution of $\bar{X}$ is exactly Gaussian; the CLT extends this asymptotically.
- Confidence intervals and hypothesis tests on the mean rely on Student's t when σ is unknown.